Agent-Based Dynamic Tariff Optimization for EV Charging Networks:

Using Large-Scale Charging Session Data

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Version 2 — corrected analysis

EXECUTIVE SUMMARY

BUSINESS CHALLENGE

Fixed EV charging tariffs ignore demand patterns, causing peak-hour congestion, uneven charger use and missed revenue.

PROPOSED SOLUTION

A framework of three agents:

- **Demand Prediction Agent** — forecasts sessions and utilization 1 hour ahead
- **Tariff Pricing Agent** — surge pricing that must not raise congestion
- **Monitoring & Learning Agent** — learns how drivers react to price and re-plans daily

KEY OUTCOMES

- **Revenue +2.2%** (₹2.33L → ₹2.39L) while congested hours **fell 4.5% → 3.6%**
- Forecast RMSE **1.55**, MAE **1.02**, R^2 **0.72** — 29% better than a naive baseline
- Price elasticity **estimated at -0.52** from real Shenzhen pricing data
- Peak (p99) utilization cut from 87.5% to 83.6%

DATA LANDSCAPE & PREPROCESSING

Data Pipeline: Raw Data → Timezone fix → Cleaning → Hourly grid → Features → AI Agents

Datasets:

ACN-Data (Caltech)

- 14,999 raw → 14,629 clean sessions
- 54 stations, 25 Apr – 15 Dec 2018
- Energy, connect / disconnect times

UrbanEV (Shenzhen)

- 247 zones, 18,061 charging piles
- 57 zones with dynamic pricing
- 19 Jun – 18 Jul 2022, 5-min data

Key Engineered Features:

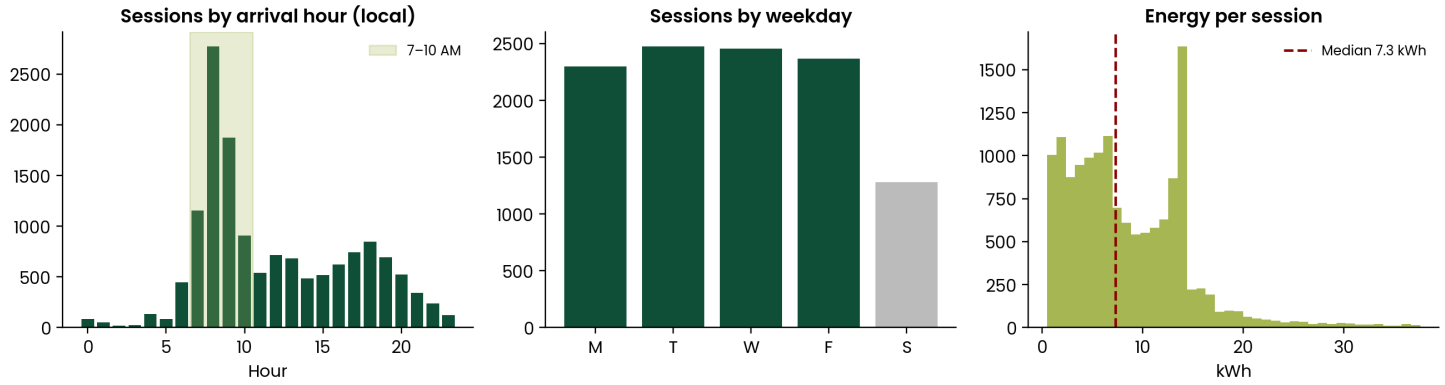
- Utilization = cars plugged in ÷ capacity
- Congestion flag (>80% utilization)
- Lags: 1h, 2h, 24h, 168h
- Rolling means: 3h, 24h, 7 days
- Hour / day cycles, holidays, trend
- Price & demand per pile (Shenzhen)

Data Preparation:

- GMT → Caltech local time (v1 peaks were 7–8 h off)
- Full hourly grid incl. zero-demand hours
- Trim top 1% energy outliers
- Shenzhen 5-min → hourly
- Capacity set from training data only (no leakage)

Key Takeaway:
Clean, time-correct hourly data with honest features for forecasting and tariff simulation.

EDA: DEMAND BEHAVIOUR & CHARGING PATTERNS

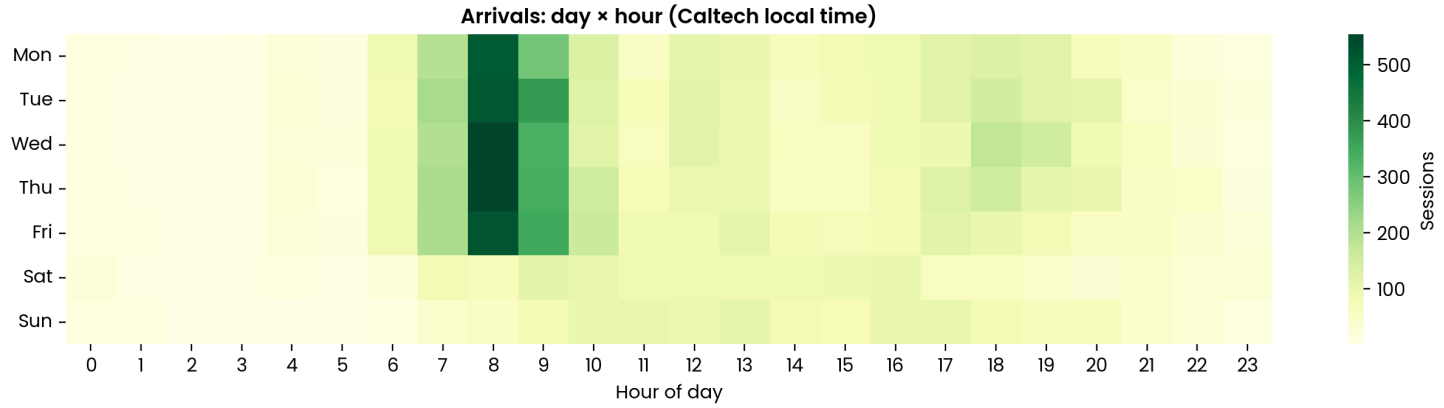


1. Morning-arrival peak

- Busiest arrival hour: **8 AM local**
- Peak slot: **Wed 8:00 (554 sessions)**
- Workplace charging: drivers plug in on arrival

2. Weekday-dominated usage

- Weekends are only **17.5%** of sessions
- Commuter-driven demand



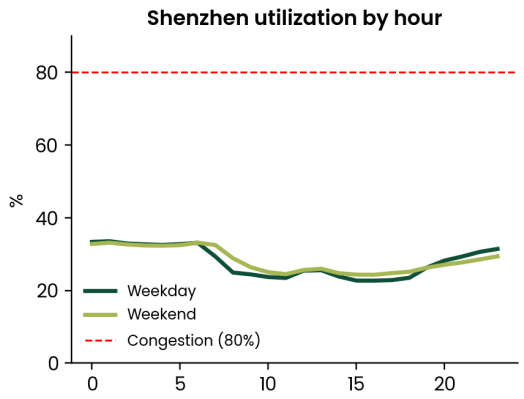
3. Energy per session

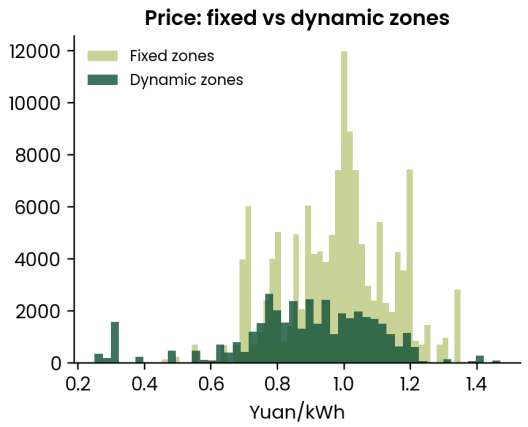
- Median ≈ **7.3 kWh**
- A few long sessions deliver very high energy

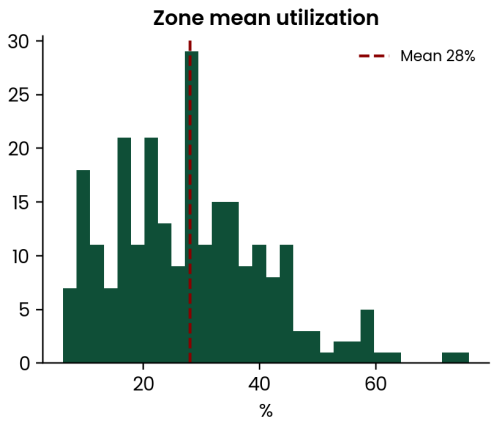
4. Charging vs parking time

- Median charging ≈ **2.2 h**
- Median parked ≈ **4.6 h**
- Cars sit idle on chargers for ~2.4 h — a bigger capacity lever than price

EDA: URBAN NETWORK (SHENZHEN)







1. Low average utilization

- Mean zone utilization ≈ **28%**
- Only **0.9%** of zone-hours above 80%

2. Night & early-morning highs

- Taxis / ride-hailing charge overnight
- Midday is the quiet period

3. A natural pricing experiment

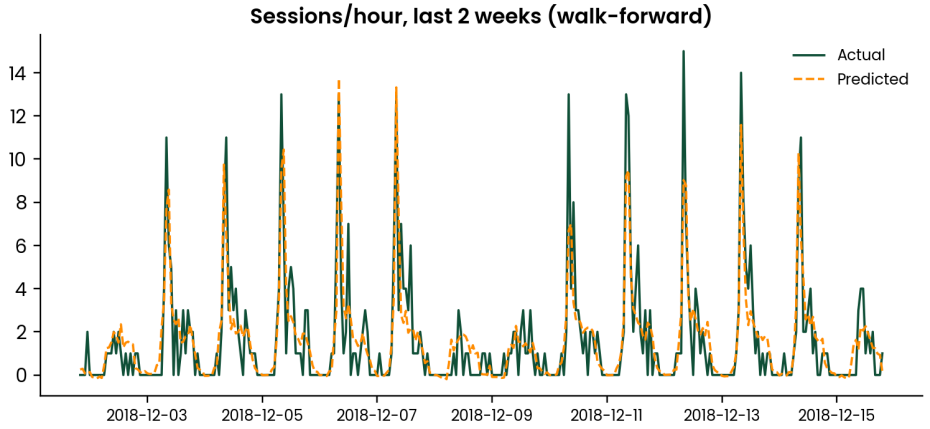
- 57 zones use dynamic pricing, the rest fixed prices
- Comparing them shows how demand reacts to price
- This is how we **estimate** elasticity (next slides) instead of assuming it

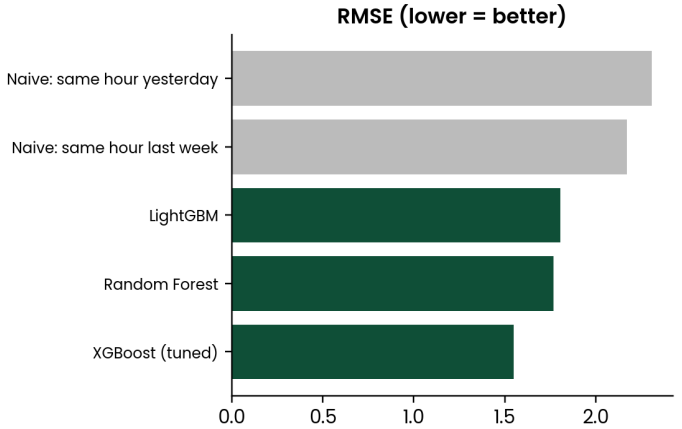
4. Station-level differences (Caltech)

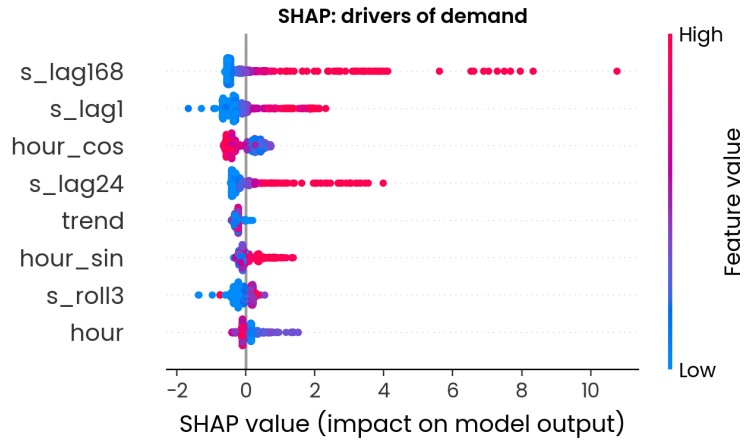
- Mean station utilization ≈ **30.9%**
- KMeans segments: 15 low, 24 medium, 15 high-use stations
- Some stations stay underused while others queue

EDA Conclusion: Demand is strongly time-dependent and uneven, and real price variation exists in the data — enough to forecast demand and measure price response.

DEMAND PREDICTION AGENT







Model	RMSE	MAE	R²
Naive: same hour last week	2.172	1.222	0.439
Naive: same hour yesterday	2.308	1.282	0.366
Random Forest	1.768	1.191	0.628
LightGBM	1.805	1.160	0.612
XGBoost (tuned)	1.549	1.016	0.715

Selected: XGBoost (tuned) — 29% lower RMSE than naive

Tuned with TimeSeriesSplit; scored walk-forward (retrained before each of 4 test blocks, 31 Oct – 15 Dec 2018)

BUSINESS IMPACT

- Forecasts sessions with MAE ≈ **1.0 per hour**
- Utilization forecast: MAE 3.5 pp, R² 0.97 — drives the tariff
- Flags congested hours with AUC 0.99

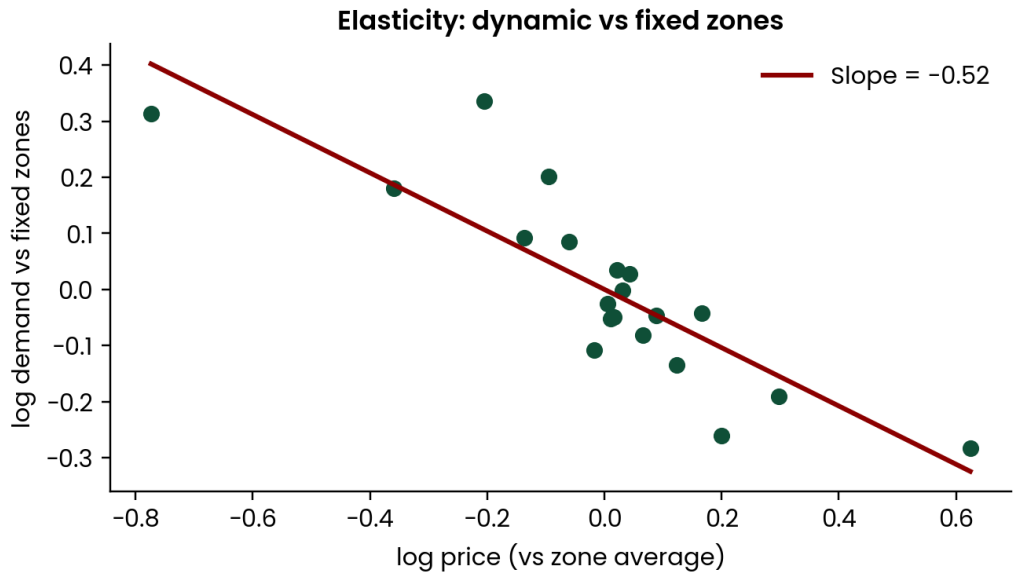
SHAP Analysis

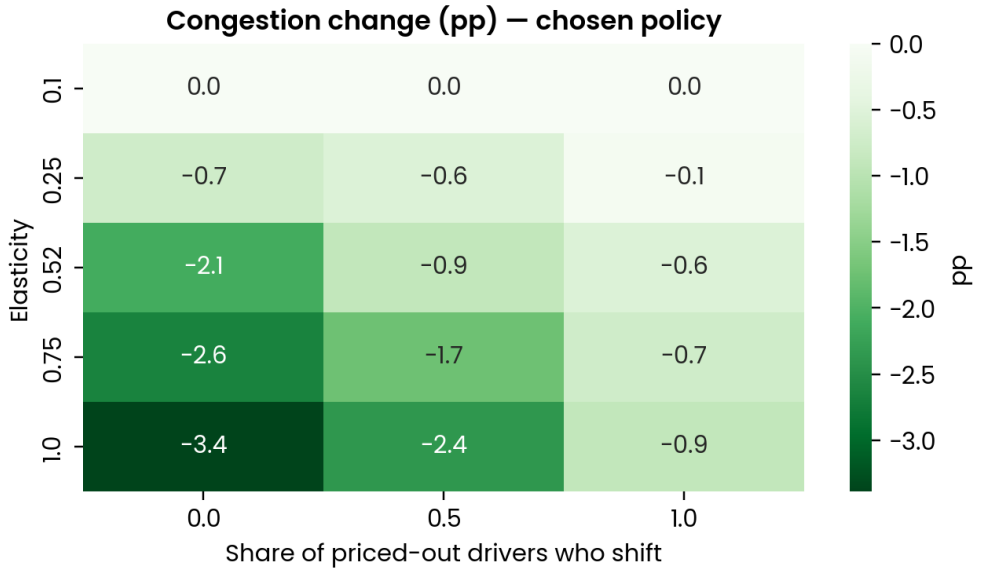
Top demand drivers:

- Same hour last week (lag-168)
- Previous hour (lag-1)
- Hour of day (cycle)
- Same hour yesterday (lag-24)
- Long-run trend

Weekly and daily cycles dominate demand.

PRICE ELASTICITY — ESTIMATED, NOT ASSUMED





Method (difference-in-differences)

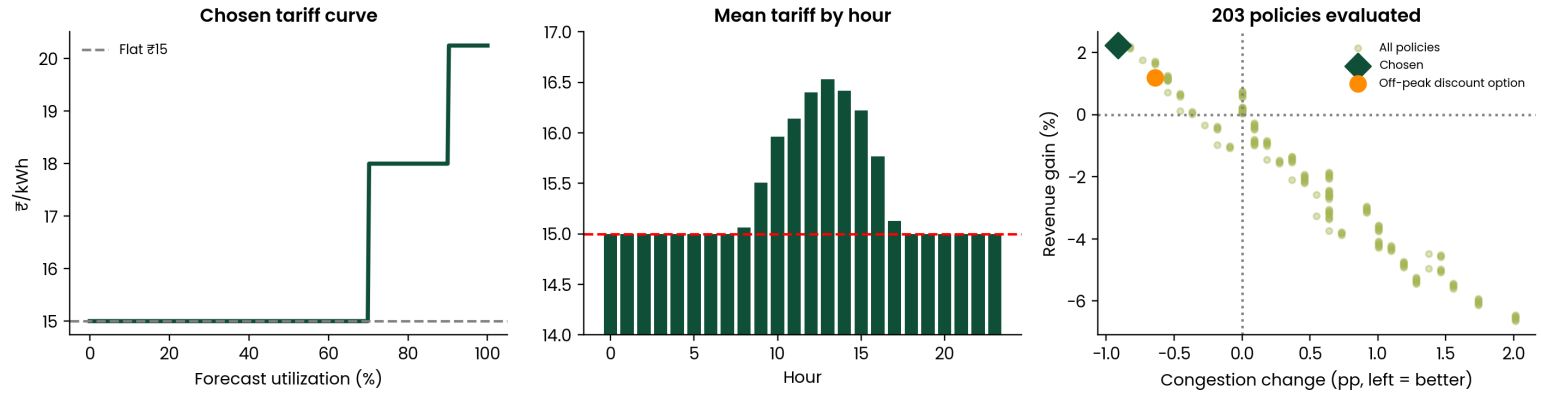
- Demand per pile in each dynamic-price zone, compared with fixed-price zones **at the same hour** — removes the normal daily cycle
- Regress on log price within each zone; slope = elasticity
- 95% interval from a day-level bootstrap

Result

- Elasticity = **-0.52** (95% CI -0.48 to -0.56): a 10% price rise cuts demand ≈ 5%
- v1 assumed -0.12, so it overstated revenue and understated congestion relief
- Caveat: operators raise prices when demand is high, so the true value is likely **more** negative
- Right chart: congestion never rises for any elasticity 0.1–1.0 or any shift share

TARIFF PRICING AGENT

Objective: Set hourly tariffs from forecast utilization to raise revenue **without** raising congestion.



KEY RESULTS

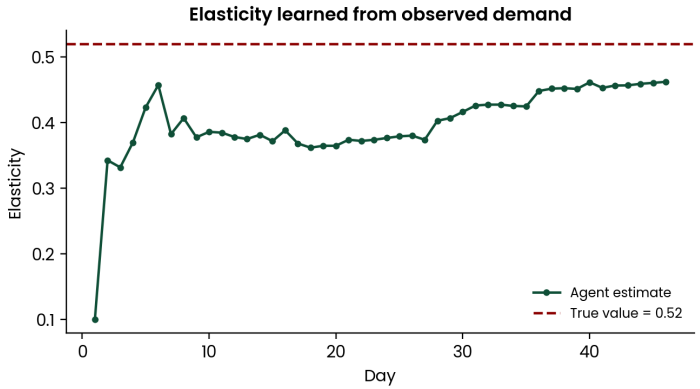
- Bands: discount <30%, base, medium surge 70–90%, high surge >90%; price cap 1.35×
- **203 policies** simulated session-by-session; **28 are win-win**
- Chosen: high **1.35×**, medium **1.20×**, discount 1.00×
- Revenue ₹2,33,361 → **₹2,38,560 (+2.2%)**
- Congested hours **4.5%** → **3.6%**; ₹/kWh 15.00 → 15.48
- Priced-out drivers: 50% shift ±2–3 h, rest leave

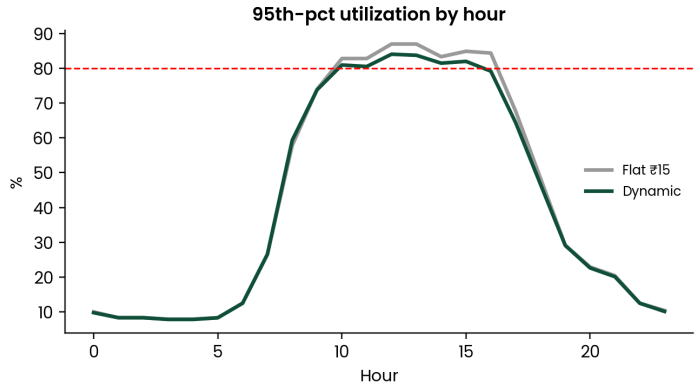
Business Insights

- Surge at midday peak both earns more **and** spreads load — no trade-off at these prices
- Off-peak discount (0.95×) lifts night sessions 2.6% but halves the revenue gain — little night demand at a workplace site
- Demand is inelastic, so without a cap revenue-max just pushes prices up — the cap is a design choice
- Segment pricing (low / medium / high-use stations) is the next step

MONITORING & LEARNING AGENT

Objective: Learn how drivers respond to price from observed demand, and re-plan the tariff every day.





Business KPI Dashboard

KPI	Flat ₹15	Dynamic	Change
Revenue (₹)	2,33,361	2,38,560	+2.2%
Mean utilization (%)	24.8	24.5	-0.3 pp
Congested hours (%)	4.5	3.6	-0.9 pp
p99 peak utilization (%)	87.5	83.6	-3.9 pp
Off-peak sessions	149	149	0.0%
₹ per kWh	15.00	15.48	+0.48

Feedback Loop

Forecast demand
↓
Plan tariff (current elasticity)
↓
Small price test (±10%)
↓
Observe demand
↓
Re-estimate elasticity

Key Takeaways

- Starts from a wrong guess (-0.10, v1's value) and learns **-0.46** vs true -0.52 (11% off) in 46 days
- Ends on the same policy an all-knowing oracle would pick
- Peak-hour 95th-percentile utilization drops toward the 80% line (right chart)
- Here the best policy is robust to elasticity, so learning mainly confirms it; it matters more for discount decisions
- Price response is simulated from the estimated elasticity — needs a live pilot to confirm

BUSINESS, OPERATIONAL & POLICY IMPLICATIONS

01

Business Impact

- Revenue +2.2% (₹2,33,361 → ₹2,38,560) **and** fewer congested hours (4.5% → 3.6%)
- Win-win in every tested elasticity scenario (-0.1 to -1.0)
- Midday surge captures peak value without pushing drivers away

02

Operational Insights

- Arrivals peak at 8 AM; chargers fill until mid-afternoon
- Cars charge 2.2 h but stay 4.6 h — idle fees could free more capacity than pricing
- Off-peak discounts don't pay at a workplace site

03

Policy Implications

- Cap surge prices: inelastic demand lets revenue-max drift up
- Combine pricing with idle fees, reservations and smart scheduling
- Next: live price pilot to validate the -0.52 elasticity

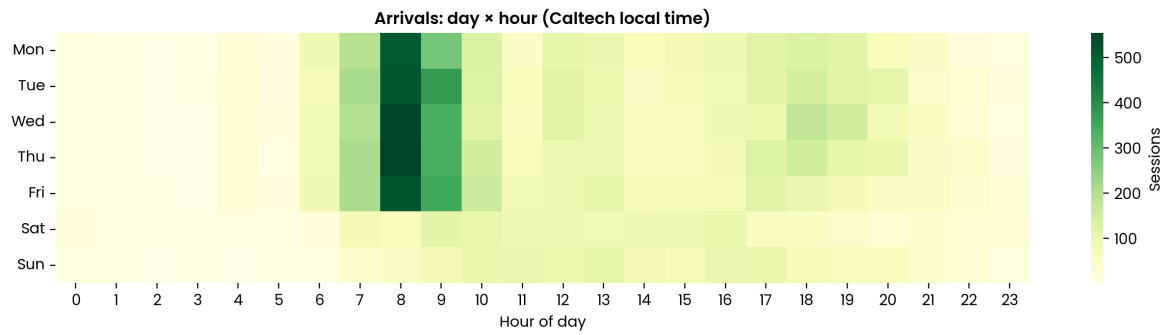
APPENDIX — TECHNICAL SUMMARY & WHAT CHANGED IN V2

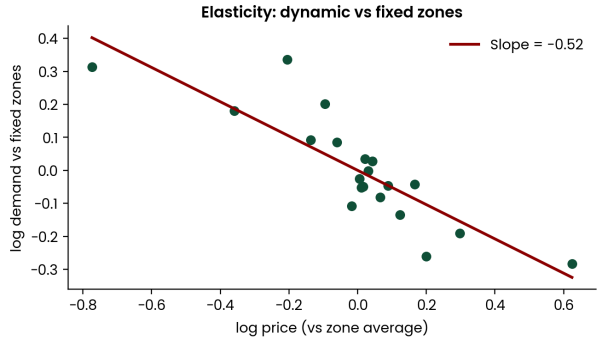
Model & Data Summary

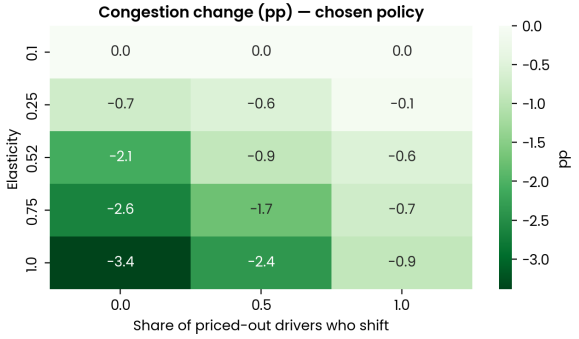
Item	Value
ACN sessions (clean)	14,629 / 54 stations
Shenzhen	247 zones / 18,061 piles
Forecast RMSE / MAE	1.549 / 1.016 per hour
Forecast R^2	0.715
vs naive (same hour last week)	-29% RMSE
Utilization forecast R^2	0.966
Congestion AUC	0.988
Elasticity (95% CI)	-0.52 (-0.48 to -0.56)
Revenue / congestion	+2.2% / -0.9 pp

v1 → v2 Fixes

- Congestion "rise" was a bug: before used **predicted**, after used **actual** utilization
- Timestamps were GMT; peak is 8 AM local, not 2–4 PM
- Zero-demand hours were missing, so lag features were wrong
- Elasticity estimated (-0.52) instead of assumed (-0.12)
- Real learning agent replaces a fixed rule sweep
- Time-series CV + walk-forward test; naive baselines added
- Hard-coded results removed; every number computed







THANK YOU